Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
Department of Computer Science and Engineering
Academic Year 2026-27

Mini Project - Weekly Progress Report

Mini Project Title: Persistent Event-Driven Memory Cache Module

Name of Students:

1. Jay Umesh Shinde (2024800115)
2. Sanchit Gaikwad (2024800029)
3. Nikhil Shelar (2024800113)

Name of the Supervisor: Prof. Suhas Kakade

Review Date	Details of Work Done and Review of Meeting	Students Signature	Remarks by Supervisor	Supervisor Signature
11/08/2026 - 19/08/2026	**Week 3 - Phase I Core Prototype and Evaluation Preparation** • Finalized the Phase I problem definition, project objectives, scope and proposed solution for a persistent in-memory cache. • Prepared the basic Java 17 project structure and core cache design for fast key/value SET, GET and DELETE operations. • Studied and prepared the Phase I Write-Ahead Logging (WAL) flow: record the update on disk before applying the corresponding change in memory. • Prepared the basic crash-recovery demonstration plan: insert data, terminate/restart the application and rebuild the cache by replaying the WAL. • Completed Phase I documentation and presentation preparation for P1 Problem Definition, P2 Existing/Related Systems, P3 Presentation and P4 Project Planning. • Prepared the roadmap for later phases; LRU/LFU benchmarking, transactions, B-tree indexing, TTL, snapshots and advanced optimizations are kept as future work.	1. _____ 2. _____ 3. _____	To be filled by Supervisor	
Next Week Plan	**Planned Work after Phase I** • Incorporate supervisor feedback from the Phase I evaluation. • Begin Phase II implementation of LRU/LFU eviction policies and benchmarking.		To be filled by Supervisor	